

WorldConsistMem: Cross-Query Consistency Evaluation for Long-Term Memory in Evolving Worlds

Anonymous authors
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Abstract

Long-term memory (LTM) systems support agents that operate over evolving interactions and changing world states. Most existing evaluations primarily report per-query or per-instance outcomes such as answer accuracy, retrieval fitness, conflict handling, or lifecycle metrics. These protocols leave open whether answers about the same persistent world jointly describe one coherent history.

We introduce an evaluation framework for cross-query world-history consistency. WorldConsistMem is a synthetic evolving-world benchmark based on dependent query bundles grounded in one deterministic gold history. Constraints are derived automatically from that history; a reader’s predicted answer set—conditioned on the memory state—is scored by machine-checkable predicates, independently of memory internals. Strict Bundle Consistency Rate (BCR) measures full constraint satisfaction; Constraint Satisfaction Rate (CSR) provides partial diagnostic resolution when BCR floors. We also report Acc-consistency gaps and ConsAcc diagnostics.

Across a corruption suite and a scaled symbolic study (102 worlds, 1,088 bundles, 7,140 queries), per-query accuracy and bundle consistency diverge: high Acc with BCR = 0 and low Acc with BCR = 1 both occur, and retrieval systems can retain high Acc while violating bundle constraints. Under a frozen Qwen reader on an H4 subset, BCR floors at 0 for all evaluated systems while mean CSR remains discriminative (about 0.26–0.33). The structured hierarchical reference attains the highest Acc (0.185) but not the strongest CSR (BM25 leads at 0.329); architecture superiority is not supported.

Contributions: (i) a formal cross-query consistency evaluation methodology; (ii) the WorldConsistMem benchmark; (iii) strict and partial consistency metrics (BCR, CSR, ConsAcc, gaps); (iv) an empirical characterization with released reference systems.

1 Introduction

Long-term memory systems increasingly support assistants and agents that operate across extended interactions and changing world states. As these systems store, update, and retrieve information over time, their answers are used for isolated facts and for decisions that depend on what is currently true, what was previously true, how transitions occurred, and which evidence supports those claims. Evaluating whether such systems preserve a usable record of an evolving world is therefore a central measurement problem for agent memory research. This paper does not claim that long-horizon memory itself is a new problem; it focuses on how consistency across dependent answers should be measured.

Most existing evaluations primarily report per-query or per-instance outcomes. Long-horizon memory benchmarks typically score answer accuracy, ability-specific accuracy, retrieval fitness, profile or lifecycle metrics, conflict handling, or temporal update success on individually graded questions (Wu et al., 2025; 2026; Hu et al., 2025; Tao et al., 2026; Xie et al., 2026; Wang et al., 2026; Hu et al., 2026; Liu et al., 2026). These protocols are informative about what a system can recover under long histories, including temporal updates

and conflict-sensitive retrieval. They leave open a complementary question: when several answers concern the same persistent world, do they jointly describe one coherent history?

Queries over one evolving world are dependent. Answers about current state, prior state, transitions, relations, provenance, and conflicts share entities, validity intervals, and evidence links. Individually plausible answers can still be mutually incompatible. For example, a system may name the correct current CEO and the correct previous CEO, yet assert an impossible transition between them, or report a provenance identifier that cannot support the stated historical claim. Per-query accuracy can remain high while the answer set fails as a world history.

We introduce an evaluation framework for cross-query world-history consistency. World-ConsistMem is an evolving synthetic-world benchmark based on dependent query bundles grounded in one deterministic gold history. After a memory system ingests an observation stream, a reader answers a bundle of interdependent questions spanning current, historical, transition, relational, provenance, and conflict-oriented slots. Constraints $\mathcal{C}$ are derived automatically from the gold history and query slots; consistency is scored by machine-checkable predicates over the predicted answer set produced by that reader conditioned on the memory state, without requiring access to memory internals. Structured reference systems appear later for evaluation only, not as the lead contribution. The research question is:

Under evolving synthetic worlds with dependent query bundles and machine-checkable cross-query constraints, do long-term memory systems that achieve non-trivial per-query accuracy also produce jointly consistent answer sets, and which partial-consistency diagnostics remain informative when strict bundle consistency floors?

WorldConsistMem operationalizes two complementary consistency metrics alongside accuracy. Bundle Consistency Rate (BCR) is strict global consistency: a bundle counts as consistent when every applicable constraint is satisfied. Constraint Satisfaction Rate (CSR) measures partial constraint satisfaction and provides diagnostic resolution when almost no bundle is fully consistent. We also report accuracy-consistency gaps (notably $\text{Gap}_{\text{query}} = \text{Acc} - \text{BCR}$) and related ConsAcc diagnostics. Under weaker readers, strict BCR may floor at zero even when systems differ in how many constraints they satisfy; CSR complements rather than replaces BCR.

Empirically, corruption and symbolic evaluations show Acc-consistency dissociation: high Acc with $\text{BCR} = 0$ and low Acc with $\text{BCR} = 1$ both occur, and scaled symbolic retrieval systems can retain high per-query Acc while violating bundle constraints. Under a Qwen reader on the frozen H4 subset, BCR floors at 0 across all evaluated systems while mean CSR remains discriminative (about 0.26–0.33). Among the evaluated Qwen reader systems, the structured hierarchical reference baseline attains the highest Acc (0.185) but not the strongest CSR (BM25 leads at 0.329). Architecture superiority is therefore not supported under the completed LLM evaluation; hierarchical H0 is released as a reference baseline for comparison.

Contributions.

1. Formal evaluation methodology for cross-query consistency of long-term memory systems over evolving worlds, using dependent query bundles and machine-checkable world-history constraints Φ .
2. WorldConsistMem, a synthetic evolving-world benchmark with shared gold histories, multi-family interdependent queries, and automatically derived constraints (102 worlds, 1,088 bundles, 7,140 queries in the scaled freeze).
3. Strict and partial consistency metrics—BCR, CSR, ConsAcc variants, and Acc-consistency gaps—that separate per-query accuracy from jointly coherent answer sets.
4. Empirical characterization of Acc-consistency dissociation under corruption, symbolic, and Qwen conditions, with released reference systems (including a structured hierarchical baseline) for reproducible comparison.

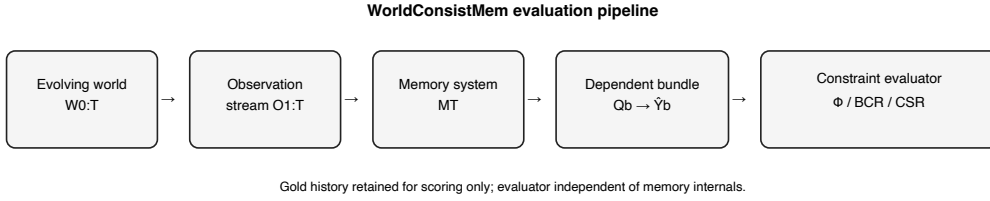


Figure 1: WorldConsistMem evaluation pipeline.

Adjacent work already separates accuracy from global logical coherence on multi-query case files (Salla et al., 2026; Chaudhry, 2026). Existing LTM benchmarks primarily emphasize long-horizon answer or stage metrics. Our residual lies at the intersection: persistent memory evaluation over evolving multi-entity worlds, where constraints arise from temporal validity, transitions, relations, provenance, and shared history. The remainder of the paper defines the evaluation object (Section 3), benchmark (Section 4), metrics (Section 5), reference systems (Section 6), setup (Section 7), results (Sections 8–9), and limitations (Section 10).

2 Related Work

We situate WorldConsistMem against long-term memory evaluation (Wu et al., 2025; 2026; Hu et al., 2025; Wang et al., 2026; Hu et al., 2026), evolving-knowledge and lifecycle benchmarks (Xie et al., 2026; Liu et al., 2026), conflict/temporal validity work (Tao et al., 2026), and adjacent cross-query logical consistency (Salla et al., 2026; Chaudhry, 2026). Existing LTM benchmarks primarily emphasize per-query or per-instance Acc (or closely related scores) under long histories, updates, conflict, or lifecycle protocols. Adjacent logical-consistency work already studies Acc–global-coherence separation for multi-query reasoning/case files, but not for persistent memory systems over evolving multi-entity gold worlds with world-history-derived constraints.

The residual is therefore not “memory over time” alone, nor “consistency” alone, but their intersection under machine-checkable world-history constraints: dependent query bundles, one shared gold history, BCR, and CSR. A detailed benchmark-by-benchmark discussion and comparison table appear in Appendix C (Table 6).

3 Cross-Query Consistency in Evolving Worlds

This section defines the evaluation object independently of any particular memory architecture. The question is whether answers produced from a memory state about one evolving world jointly describe that world. Figure 1 summarizes the evaluation pipeline from evolving world to constraint scoring.

3.1 Evolving world and observations

Let $W_0, \dots, W_T$ be a sequence of structured world states over a fixed entity vocabulary (persons, organizations, projects, documents, roles, ownership, and related relations). An event sequence $E_{1:T}$ drives deterministic transitions

$$W_t = \text{Apply}(W_{t-1}, E_t). \quad (1)$$

Each event carries a timestamp, typed payload, and provenance identifier. Validity intervals for roles and attributes are induced by start, end, transfer, and update events. The gold history $(W_{0:T}, E_{1:T})$ is retained for scoring and is never exposed directly to the memory system.

The system instead receives an observation stream $O_{1:T}$ derived from $E_{1:T}$ (canonical text and structured fields, optionally with distractors). After ingesting $O_{1:T}$, the system holds

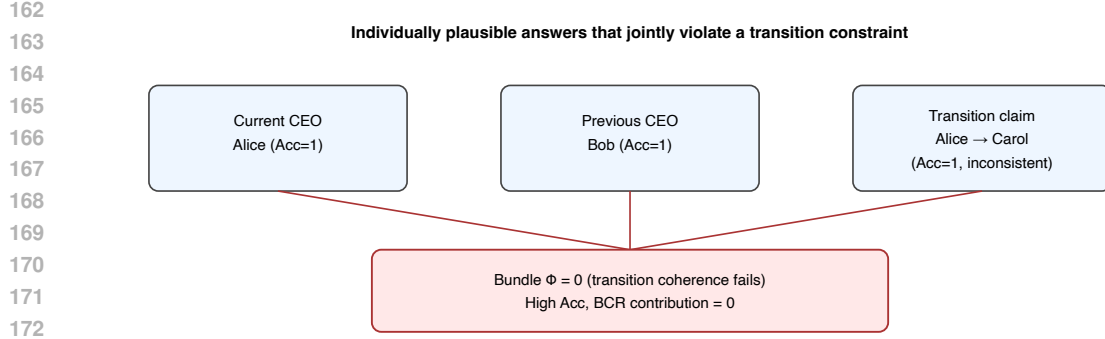


Figure 2: Individually plausible answers that jointly violate a transition constraint.

an implementation-specific memory state M_T . Because the reader conditions its predictions on M_T , differences in what each memory system retains can change $\hat{Y}_b$ and therefore affect Φ and the consistency rates defined in Section 5.

3.2 Dependent query bundles

A query bundle $Q_b = \{q_1, \dots, q_{n_b}\}$ is a finite set of queries about the same world and focal entity or relation cluster (for example, the CEO lineage of one organization, or the ownership chain of one project). Bundle members are intentionally dependent: they ask for current state, historical state, transitions, temporal order, provenance, multi-hop relations, contradictions, counterfactuals, or conflict validity over shared entities and times.

Gold answers $Y_b^* = \{y_i^*\}$ are derived only from $(W_{0:T}, E_{1:T})$. Predictions $\hat{Y}_b = \{\hat{y}_i\}$ are produced by a reader conditioned on M_T (and optional retrieved evidence). Per-query accuracy compares $\hat{y}_i$ to y_i^* under structured equality. Accuracy alone does not ask whether the predicted answers are jointly coherent.

3.3 Automatically derived constraints and Φ

From the gold history and the bundle’s query slots we derive a finite set of machine-checkable constraints $\mathcal{C}_b$ over predicted answers (and, where applicable, predicted provenance). Constraint families include uniqueness, temporal order, transition coherence, relational agreement, provenance support, multi-hop composition, and contradiction/conflict validity. Constraints are predicates on $\hat{Y}_b$; they do not require re-running the memory system.

Define the consistency predicate

$$\Phi(\hat{Y}_b; \mathcal{C}_b) = \begin{cases} 1 & \text{if every } c \in \mathcal{C}_b \text{ holds on } \hat{Y}_b, \\ 0 & \text{otherwise.} \end{cases} \quad (2)$$

When $\mathcal{C}_b = \emptyset$, we take $\Phi = 1$ (vacuous consistency). Strict bundle consistency is $\Phi = 1$; Section 5 defines aggregate rates and partial satisfaction.

3.4 Accuracy versus world consistency

Two failure modes motivate the separation:

High accuracy, inconsistent answers. A system may correctly name the current CEO and the previous CEO while asserting a transition time or provenance that cannot reconcile those two facts. Per-query Acc can remain high while $\Phi = 0$ (Figure 2).

Low accuracy, internally consistent alternative world. A system may answer an entire bundle with a coherent but wrong ownership lineage (consistent among themselves, wrong

relative to gold). Then Acc is low while $\Phi = 1$. Consistency is therefore not a proxy for correctness; it measures joint coherence of the answer set as a candidate world description.

The evaluation object is this Acc- Φ distinction for memory systems over evolving multi-entity worlds. Adjacent multi-query logical consistency work studies Acc versus global coherence for reasoning/SAT settings (Salla et al., 2026; Chaudhry, 2026); the present object is constraint checking of memory answers against world-history-derived predicates, not SMT satisfiability of logical commitments alone.

3.5 Implementation independence

Any system that maps observation streams to a memory state and answers queries—flat stores, retrieval pipelines, structured hierarchical stores, or LLM readers over retrieved evidence—can be scored with the same $(Q_b, Y_b^*, \mathcal{C}_b)$ protocol. The formal object does not privilege any particular multi-store layout.

4 WorldConsistMem Benchmark

WorldConsistMem is a deterministic synthetic benchmark that instantiates Section 3. It is designed so that consistency constraints are automatically derived from a shared gold history, and so that bundles are not bags of unrelated questions.

4.1 Entities, events, and validity

Worlds contain typed entities (persons, organizations, projects, documents, locations) and relations (employment/CEO roles, ownership, membership, document versions, dependencies). An event grammar emits start, end, transfer, update, and related lifecycle events. Role and attribute assertions induce validity intervals. Each event carries a provenance identifier used by provenance queries and provenance constraints.

Generation is seed-deterministic. Given a seed and tier, the world, event log, observation stream, bundles, gold answers, and applicable constraints are fixed.

4.2 Observation stream

Systems never read gold W_t directly. They ingest an observation stream: structured fields plus canonical text (and, in smoke settings, paraphrases/distractors as configured). Observations are the only channel through which event information enters memory.

4.3 Query-bundle generation and families

For each focal lifecycle (for example, a CEO succession or project ownership chain), the generator emits a bundle whose queries share that focus and therefore share entities, times, and supporting events. Query families include: current state; historical state; transition; temporal order; provenance; multi-hop; contradiction; counterfactual; and conflict validity.

Why bundles are not unrelated questions. Bundle members are linked by construction: answering “who is CEO now,” “who was CEO before,” “when did the change occur,” and “what evidence supports the current role” refers to one succession. Constraints such as transition coherence and provenance support are meaningful only because those answers must agree. A shuffled pool of independent factual questions would not induce the same $\mathcal{C}_b$.

4.4 Constraint derivation

Constraints are compiled from gold history and query slots (uniqueness of exclusive roles, temporal order of events, transition payload agreement with current/historical answers, relational closure for multi-hop items, provenance alignment, contradiction/conflict rules). The evaluator applies $\mathcal{C}_b$ to predictions without consulting the memory implementation’s internals.

Table 1: WorldConsistMem dataset statistics (scaled freeze).

Quantity	Value
Worlds	102 (34 / 34 / 34)
Bundles	1,088
Queries	7,140
Intermediate / complex bundles	340 / 748
Mean events (small / medium / large)	35 / 97 / 215
H4 LLM subset	18 worlds, 192 bundles, 1,260 queries

4.5 Dataset scale and roles

Dataset scale is summarized in Table 1. The scaled benchmark (seed 100) contains 102 worlds (34 small / 34 medium / 34 large), 1,088 bundles, and 7,140 queries. Tiers differ in event volume (mean events per world: 35 / 97 / 215). Bundle difficulty labels (intermediate, complex) are structural annotations for stratified reporting.

The smoke benchmark confirms gold metrics ($\text{Acc} = \text{BCR} = \text{ConsAcc} = 1$) and the corruption suite that separates Acc from Φ . Scaled evaluation reuses the same metric and constraint path; smoke is the regression lock, scaled is the primary empirical setting.

The H4 LLM subset uses the first 6 worlds per tier (18 worlds), 192 bundles, and 1,260 queries, used only for the local Qwen reader study (Sections 7–8). Symbolic scaled results use the full 102-world set unless noted.

Query-family counts (scaled): current/historical/transition/multi-hop/provenance = 1,088 each; temporal_order/conflict_validity = 442; contradiction/counterfactual = 408.

4.6 Scope statement

WorldConsistMem is a controlled synthetic-world benchmark. Its purpose is to make Acc and joint world-answer consistency separately measurable under identical gold histories and constraints. Ecological and deployment limits are stated in Section 10.

5 Evaluation Metrics

Metrics are reported jointly. We do not collapse evaluation into a single headline score. Strict BCR remains the primary consistency rate; CSR and partial ConsAcc are complementary diagnostics, especially when BCR floors.

5.1 Per-query accuracy

For query q_i with gold y_i^* and prediction $\hat{y}_i$,

$$\text{Acc}_i = \mathbf{1}[\hat{y}_i \equiv y_i^*] \quad (3)$$

under structured equality (IDs, times, relation slots). Mean query accuracy Acc averages Acc_i over all scored queries. We also report accuracy by query family.

5.2 Bundle Consistency Rate (BCR)

Let B be the set of bundles. With Φ as in Section 3,

$$\text{BCR} = \frac{1}{|B|} \sum_{b \in B} \mathbf{1}[\Phi(\hat{Y}_b; \mathcal{C}_b) = 1]. \quad (4)$$

BCR is strict: a single violated applicable constraint yields $\Phi = 0$ for that bundle. Under weak readers, BCR can floor at zero even when systems differ in how many constraints they satisfy.

5.3 Constraint Satisfaction Rate (CSR)

For bundle b ,

$$\text{CSR}_b = \frac{\#\{\text{satisfied constraints in } b\}}{\#\{\text{applicable constraints in } b\}}, \quad (5)$$

with $\text{CSR}_b = 1$ when $\mathcal{C}_b = \emptyset$. Equivalently, BCR is the fraction of bundles with $\text{CSR}_b = 1$. We report mean and median CSR, CSR distributions, CSR by constraint family, and CSR by difficulty/tier.

We also report the mean/median number of violated constraints and normalized violations $1 - \text{CSR}_b$ when $|\mathcal{C}_b| > 0$. CSR does not replace BCR. It provides resolution when almost no bundle is fully consistent.

5.4 Consistency-aware accuracy

Strict ConsAcc. Fraction of bundles that are fully correct ($\text{Acc}_i = 1$ for all $i \in Q_b$) and $\Phi = 1$.

Relaxed / partial ConsAcc. (i) Relaxed ConsAcc: mean query Acc on the bundle $\geq \tau$ and $\Phi = 1$ (default $\tau = 0.8$). (ii) Partial ConsAcc grid $\text{PConsAcc}_{\tau, \tau_c}$: fraction of bundles with mean query Acc $\geq \tau$ and $\text{CSR}_b \geq \tau_c$, for predeclared $\tau \in \{0.5, 0.8\}$ and $\tau_c \in \{0.8, 0.9\}$. We report the full grid; we do not cherry-pick a favorable cell as a primary claim.

5.5 Accuracy-consistency gaps

Primary gap:

$$\text{Gap}_{\text{query}} = \text{Acc} - \text{BCR}. \quad (6)$$

Secondary gap:

$$\text{Gap}_{\text{bundle}} = \text{Acc}_{\text{bundle}} - \text{ConsAcc}_{\text{strict}}, \quad (7)$$

where $\text{Acc}_{\text{bundle}}$ is the fraction of bundles with all answers individually correct. Large positive $\text{Gap}_{\text{query}}$ indicates high per-query Acc relative to strict joint consistency.

5.6 Family diagnostics

We report temporal and provenance violation rates; constraint-family violation rates (fraction of bundles with ≥ 1 failure in each family); high-accuracy inconsistent rate (mean Acc ≥ 0.8 and $\Phi = 0$); and abstention/malformed rates for LLM readers.

5.7 Statistical protocol

Primary units for confidence intervals are worlds, not queries. We report world-level bootstrap 95% CIs for Acc, BCR, ConsAcc, and CSR where applicable. Queries within a bundle are not treated as independent samples.

6 Evaluated Memory Systems

Systems are reference implementations for evaluation. We describe them factually. We do not claim architectural superiority for any design, including the structured hierarchical reference.

6.1 Oracle and flat stores

- Oracle / gold. Predictions equal gold answers (validation upper bound; $\text{Acc} = \text{BCR} = 1$ on well-formed bundles).
- Flat append-only (unconstrained). All observations retained; reader consults the full store.
- Latest-only. Retain only the most recent observation(s) under a simple currency rule; older evidence is discarded.

- Capacity-limited flat. Fixed logical-record budget with eviction: recency, compact padding (drop low-priority padding/status noise first), or reservoir sampling.

6.2 Retrieval baselines

- Recency- k . Retrieve the k most recent observations.
- BM25- k . Lexical retrieval of top- k observations for the query text.

Reported operating points include $k \in \{3, 8\}$ under the frozen scaled protocol.

6.3 Structured hierarchical reference (H0)

H0 is a structured lifecycle-aware store with multiple internal components (working/session-style buffers, long-term assertions with validity, graph edges, provenance, archive/conflict bookkeeping). At query time it returns an evidence pack under a retrieval budget (e.g., ret8 at capacity 150).

Role in the paper. H0 is a reproducible structured baseline, not presented as a superior architecture. Symbolic matched-budget comparisons and the Qwen reader study report Acc and consistency for H0 under the WorldConsistMem metrics. Completed Qwen results do not support a consistency advantage for H0 (Section 8).

6.4 Readers

- Symbolic / deterministic reader. Exact structured matching over retrieved or stored records (primary scaled symbolic tables).
- Local LLM reader. Qwen2.5-3B-Instruct-4bit with frozen prompt, temperature 0, max tokens, JSON early-stop, and parser (H4 subset).

Evidence packs are isolated per query under the frozen H4 protocol so that cross-query leakage through the prompt is not an uncontrolled confound.

6.5 Fairness limitations (explicit)

Comparisons are informative but not perfectly matched (internal structure, capacity accounting for H0 metadata, retrieval-budget identity, and compact-flat padding eviction). Full fairness caveats appear in Appendix D. These limits bound interpretation: gaps support metric claims; they do not license claims of architectural superiority.

7 Experimental Setup

All results in Sections 8–9 use frozen artefacts. No weights, prompts, metrics, or datasets were altered for this manuscript.

Symbolic reader. Full scaled WorldConsistMem (102 worlds, 1,088 bundles, 7,140 queries), including flat, retrieval, capacity-matched, H0, and selective ablation systems.

Qwen reader (H4 Option A). Qwen2.5-3B-Instruct-4bit on 18 worlds / 192 bundles / 1,260 queries for B1 compact, B3 recency- k 8, B4 BM25- k 8, and H0 (cap150, ret8). Per-query isolated evidence packs; temperature 0; max tokens 48; frozen parser. Phi was not used for Option A.

Validation and statistics. Smoke corruptions confirm Acc and Φ can move in opposite directions (Section 8). World-level bootstrap 95% CIs; symbolic and Qwen tables reported separately. CSR/PConsAcc recomputed offline from frozen predictions. Extended protocol details appear in Appendix A.

Table 2: Corruption-suite validation (smoke; frozen).

Corruption	Acc	BCR	Gap _{query}
high_acc_inconsistent	0.849	0.000	0.849
wrong_provenance	0.849	0.000	0.849
consistent_wrong_world	0.394	1.000	−0.606
transition_inconsistency	0.939	0.600	0.339

Table 3: Symbolic scaled results (102 worlds; frozen). Acc, BCR, CSR rounded half-up to three decimals from frozen point estimates.

System	Acc	BCR	Gap _{query}	mean CSR
B3 recency- <i>k</i> 8	0.567	0.063	0.504	0.507
B4 BM25- <i>k</i> 8	0.929	0.591	0.338	0.881
B1 compact (cap150)	1.000	1.000	0.000	1.000
H0 LTKM (cap150, ret8)	0.976	0.844	0.132	0.973

8 Results

We organize results around five questions. Symbolic and Qwen conditions are never pooled.

8.1 Are accuracy and consistency distinct?

Corruption suite (smoke). Gold predictions attain $\text{Acc} = \text{BCR} = \text{ConsAcc} = 1$. Controlled corruptions separate the metrics (Table 2). High Acc with $\text{BCR} = 0$ and low Acc with $\text{BCR} = 1$ both occur. This supports that Φ is not redundant with Acc.

8.2 Does the distinction persist at scale?

Symbolic scaled (102 worlds). Retrieval baselines show large Acc–BCR gaps (Table 3; Figure 3, left). B4 retains high Acc while BCR remains materially lower ($\text{Gap}_{\text{query}} = 0.338$). The Acc–consistency distinction survives scaling under the deterministic reader. World-level bootstrap CIs are reported in the frozen statistical analysis artefacts.

8.3 Which systems show the largest gaps?

Under the symbolic reader, recency retrieval exhibits the largest $\text{Gap}_{\text{query}}$ among the headline retrieval points (0.504). BM25 narrows but does not close the gap. Compact flat at capacity 150 saturates Acc and BCR in this generator regime (fairness caveat: padding eviction can preserve focal lifecycle evidence; Section 6). H0 at cap150/ret8 achieves high Acc and high BCR/CSR, but does not beat compact flat on BCR (ΔBCR vs compact = −0.156 at this point; matched lift vs compact fails the frozen gate). H0 does beat matched recency flat on BCR in the frozen capacity study; that is a matched-baseline comparison, not a claim of architectural superiority over all flat stores.

Selective ablations (archive, temporal-validity, graph, conflict) produce family-level Acc drops in the frozen ablation table; they are diagnostic of component–family links under the symbolic reader, not evidence that a multi-component store is necessary. Stratified Acc/BCR by difficulty and tier are shown in Figure 5 (Appendix E).

8.4 What happens under the Qwen reader?

H4 subset; frozen Qwen Option A. All four systems have $\text{BCR} = 0$. Acc remains ordered (Table 4; Figure 3, right). Because the constraint suite is satisfiable for gold/oracle predictions and yields non-trivial BCR on the deterministic symbolic reader (Section 4 smoke

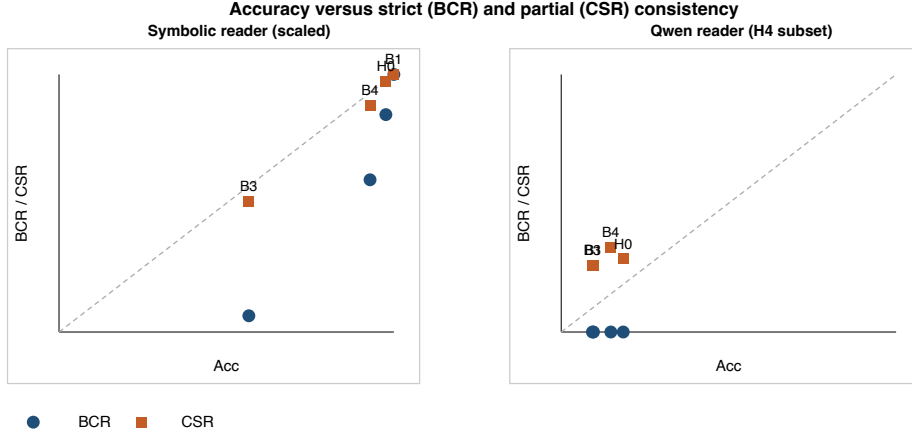


Figure 3: Acc versus BCR and CSR under symbolic and Qwen readers.

Table 4: Qwen H4 Option A results (18 worlds, 192 bundles; frozen). Values rounded half-up to three decimals. All PConsAcc grid cells at $(\tau, \tau_c) \in \{0.5, 0.8\} \times \{0.8, 0.9\}$ are 0.

System	Acc	BCR	mean CSR	Abs	Mal
B1 compact	0.096	0.000	0.258	0.728	0.007
B3 recency- k 8	0.093	0.000	0.258	0.728	0.007
B4 BM25- k 8	0.148	0.000	0.329	0.579	0.001
H0 LTKM	0.185	0.000	0.284	0.483	0.029

confirmation and Section 8 symbolic results), the Qwen BCR floor reflects difficulty in producing jointly constraint-satisfying answers under this frozen reader/parsing pipeline, not an intrinsically inconsistent constraint definition.

Reading.

- In this Qwen setting, H0 has the highest Acc and lower abstention than flats/recency.
- B4 has the highest CSR.
- H0’s CSR is above B1/B3 (+0.026) but below B4 (−0.045).
- No architecture consistency advantage is supported under Qwen: BCR ties at 0; CSR does not rank H0 highest.
- Acc gains must not be converted into consistency claims.

8.5 Does CSR provide resolution when BCR floors?

Yes, within limits. With $\text{BCR} = 0$ for all Qwen systems, mean CSR still separates B4 (0.329) from B1/B3 (0.258) and H0 (0.284). Mean violations per bundle remain high (≈ 3.3 – 3.6). Acc–CSR Pearson correlation is positive for B1/B3/B4 (≈ 0.30 – 0.43) but near zero/slightly negative for H0 (−0.04), consistent with H0 answering more questions without proportional joint constraint satisfaction.

Conclusion of Section 8. Acc and consistency are empirically distinct; the distinction persists symbolically at scale; under Qwen, strict BCR floors while CSR remains weakly diagnostic; the structured reference H0 is not supported as a consistency winner under Qwen.

Table 5: Qwen violation multiplicity (H4; frozen).

System	Mean viol./bundle	Frac. exactly one	Frac. multiple
B1 compact	3.59	0.031	0.969
B3 recency	3.59	0.031	0.969
B4 BM25	3.33	0.177	0.823
H0 reference	3.50	0.000	1.000

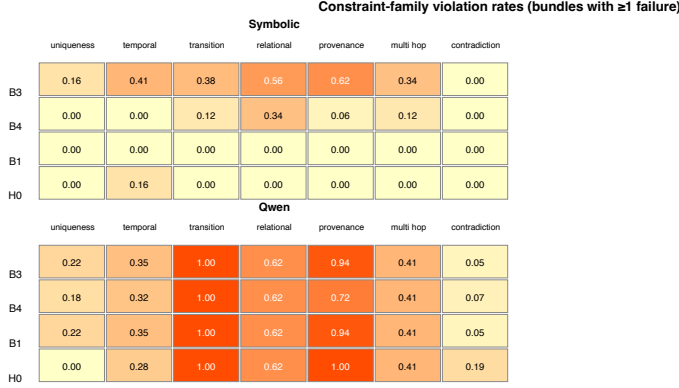


Figure 4: Constraint-family violation rates under symbolic and Qwen readers.

9 Consistency Failure Analysis

This section localizes failures under the frozen Qwen condition and contrasts them with symbolic error patterns. Counts refer to bundles with at least one violation in a family unless noted.

9.1 Qwen: widespread multiple violations

For all four Qwen systems, transition constraints fail on 192/192 bundles in the H4 subset. Failures are not explained by a single rare constraint on an otherwise consistent bundle:

BCR = 0 under Qwen therefore reflects widespread multi-constraint inconsistency, not one recurring isolated miss.

9.2 Constraint families (Qwen)

Family hit counts (bundles with ≥ 1 failure) and violation rates are reported in Figure 4 and Table 7; query-family Acc appears in Table 8 (Appendix E). Transition and provenance dominate. H0 reduces uniqueness hits to zero in this snapshot but increases contradiction hits and still fails every transition constraint. B4 reduces provenance hits relative to B1/B3, aligning with its higher CSR.

9.3 Abstention and malformed output

Qwen abstention is high for B1/B3 (0.728), lower for B4 (0.579) and H0 (0.483). Malformed rates remain low for flats/BM25 (≤ 0.007) and higher for H0 (0.029). Lower abstention with higher Acc under H0 does not restore BCR; some additional answered queries still violate joint constraints (near-zero Acc-CSR correlation).

9.4 Evidence availability versus reader composition

Symbolic BM25 and H0 can place correct evidence in scope and still show Acc–BCR gaps; under Qwen, error decomposition emphasizes reader composition failures even when retrieval improves Acc. Frozen H4 notes include non-zero reader-failure-on-correct-evidence rates. The methodological implication suggested by the decomposition is consistent with the observed pattern that improving evidence can shift Acc without necessarily restoring $\Phi = 1$ under this frozen evaluation pipeline.

9.5 Symbolic contrast (brief)

Under the deterministic reader, family Acc and BCR are much higher; gaps concentrate on retrieval/capacity regimes (e.g., recency) rather than universal transition collapse. Ablations link archive/temporal/graph components to family Acc drops. This contrast indicates that Qwen BCR flooring is reader-regime-specific, not a claim that the constraint suite is unsatisfiable in principle (gold and strong symbolic systems achieve BCR = 1).

10 Discussion and Limitations

10.1 What the residual supports

WorldConsistMem’s residual is methodological: dependent bundles over one gold evolving world, machine-checkable cross-query constraints, BCR, and CSR make Acc and joint world-answer consistency separately measurable for memory systems. Completed evidence is consistent with these quantities diverging under corruption, symbolic retrieval baselines, and a local Qwen reader.

10.2 Strict versus partial consistency

BCR is the right strict rate for “fully coherent bundle.” When BCR floors, CSR and violation distributions remain informative and should be reported. Partial ConsAcc grids should be reported in full; they are not a license to replace BCR with a favorable threshold cell.

10.3 Synthetic-world and template limits

Worlds are synthetic; observations are templated; ecological validity for open-domain conversation is not claimed. Difficulty labels are structural. Results characterize controlled WorldConsistMem regimes, not production assistants.

10.4 Reader limits

The LLM condition uses a single local 3B-class Qwen model with frozen short-form JSON decoding. Stronger models, cloud APIs, and Phi were not part of the frozen Option A decision. Low Acc and high abstention limit how far Qwen tables can speak to “human-level” memory behavior; they remain valid for the Acc≠consistency and BCR-floor/CSR questions under this reader.

10.5 Compact-flat strength and fairness

Under symbolic matched budgets, compact flat can match or exceed H0 on BCR by discarding padding while retaining focal events; capacity accounting taxes H0 metadata (Section 6). Structure is a reference design under test, not a proven necessity.

10.6 Negative architecture result under Qwen

The hierarchical reference does not earn a consistency win under Qwen (BCR = 0 for all; CSR highest for BM25). Within the frozen Qwen evaluation, Acc improvements and

structural memory do not reliably yield joint coherence. This negative result is reported as a limitation of architecture claims under WorldConsistMem, not as a contribution headline.

10.7 Adjacent logical-consistency literature

As discussed in Section 2, cross-query logical consistency work (Salla et al., 2026; Chaudhry, 2026) already separates Acc from global coherence for multi-query reasoning, but not for persistent memory systems over evolving multi-entity gold worlds with world-history-derived constraints.

Coverage bounds for the frozen evaluation are summarized in Table 9 (Appendix E).

10.8 Proper scope of conclusions

Safe conclusions: Acc $\neq$ consistency for this benchmark; BCR can floor; CSR can still separate systems; architecture superiority is unsupported under the frozen Qwen protocol.

Unsafe conclusions: real-world generality; state-of-the-art rankings; necessity of hierarchy; BCR sufficiency alone in low-Acc regimes without CSR.

11 Conclusion

We presented WorldConsistMem, a benchmark and evaluation methodology for cross-query consistency of long-term memory systems over evolving synthetic worlds. Dependent query bundles, a shared gold history, and machine-checkable world-history constraints make per-query Acc and jointly coherent answer sets separately measurable via BCR, CSR, ConsAcc, and Acc-consistency gaps.

Frozen evidence supports four scoped findings. First, Acc and consistency are empirically distinct under corruption and at symbolic scale. Second, high-Acc retrieval regimes can still violate temporal, relational, transition, and provenance constraints across dependent queries. Third, under the completed Qwen H4 protocol, strict BCR floors while CSR remains weakly discriminative. Fourth, the structured hierarchical reference is not supported as a consistency winner under Qwen.

These conclusions are limited to synthetic worlds, exact-match Acc, the frozen constraint library, and a single local 3B-class reader. They do not establish real-world generality, state-of-the-art rankings, or necessity of hierarchical memory. Within those bounds, WorldConsistMem operationalizes a residual measurement object at the intersection of persistent memory evaluation and world-history-derived cross-query constraints.

Ethics Statement

WorldConsistMem uses fully synthetic worlds, templated observations, and machine-checkable constraints. No human subjects, personal data, or deployed decision systems are involved. We see no direct ethical risk beyond the usual dual-use considerations of publishing evaluation methodology for agent memory systems. Broader societal claims about real-world AI accountability are not made from this synthetic evidence.

Reproducibility Statement

We release the seed-deterministic scaled dataset (seed 100; 102 worlds / 1,088 bundles / 7,140 queries), frozen symbolic and Qwen predictions, metric definitions (Acc, BCR, CSR, ConsAcc, gaps), decoding freeze for the Qwen reader, and reference system implementations needed to reproduce Tables 2–9 and Figures 1–4. Partial consistency diagnostics are recomputed offline from frozen predictions without new inference. Phi was not used for the Option A decision reported here.

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A Additional Reproducibility Details

Dataset. Scaled WorldConsistMem is generated with seed 100. Smoke validation confirms gold $\text{Acc} = \text{BCR} = \text{ConsAcc} = 1$ and $\text{Acc} - \Phi$ separation under controlled corruptions.

Readers. Symbolic exact-match scoring is used for the full scaled study. The LLM condition uses Qwen2.5-3B-Instruct-4bit with frozen short-form JSON decoding (temperature 0; max tokens 48) on the H4 subset (18 worlds / 192 bundles / 1,260 queries).

Systems. Headline reference systems: B1 capacity-limited compact flat; B3 recency retrieval; B4 BM25 retrieval; H0 structured hierarchical reference. Ablations and capacity sweeps are reported in the released artefacts.

Metrics. Primary consistency rate: BCR. Complementary: CSR, ConsAcc / PConsAcc grid, Gap_{query}, family violation rates. World-level bootstrap 95% CIs accompany Acc/BCR/CSR where released.

B Artifact Checklist

Released artefacts accompanying this submission include: dataset generators and manifests; symbolic system results and predictions; Qwen H4 system/query-family/constraint-family results; partial-consistency recomputes; corruption-suite metrics; and figure sources corresponding to Figures 1–4. Anonymized supplementary material is provided with the OpenReview submission; identifying repository URLs are omitted from this PDF.

C Extended Related Work

C.1 Long-term memory evaluation

LongMemEval evaluates chat assistants on long-term interactive memory through five ability axes—information extraction, multi-session reasoning, temporal reasoning, knowledge updates, and abstention—with answer accuracy as the primary metric under long-history and long-context protocols (Wu et al., 2025). LongMemEval-V2 extends the agenda toward “experienced colleague” competence, adding dynamic state tracking, workflows, and premise awareness, again with per-question accuracy (and latency-oriented reporting) as the headline outcomes (Wu et al., 2026). MemoryAgentBench evaluates agent memory via incremental multi-turn interactions across competencies such as accurate retrieval, test-time learning, long-range understanding, and selective forgetting / FactConsolidation after edits, scored primarily by task or answer accuracy (Hu et al., 2025). EverMemBench targets long-horizon collaborative, multi-party dialogues and reports fine-grained recall, memory awareness, and profile understanding under QA accuracy protocols (Hu et al., 2026). EvoMemBench organizes agent memory from a self-evolving perspective with a scope×content taxonomy and reports task success or accuracy (and efficiency) across methods compared with long-context baselines (Wang et al., 2026). Taken together, this line provides mature protocols for long-horizon Acc and ability factorization; the measurement gap we target is joint coherence of dependent answers, not another Acc axis.

These benchmarks primarily evaluate individual query or task outcomes. WorldConsistMem does not replace ability Acc suites. It additionally evaluates whether answers within a dependent bundle jointly satisfy constraints derived from one world history. The evaluation unit is the interdependent answer set $\hat{Y}_b$ under $\Phi(\hat{Y}_b; \mathcal{C}_b)$, not a single graded QA item.

C.2 Dynamic and evolving knowledge

Evolving worlds are not novel by themselves. DynamicMem constructs long-horizon personal memory in real-world settings and evaluates temporal checkpoint reconstruction and related state-tracking accuracy for evolving user attributes, habits, and preferences (Xie et al., 2026). WorldMemArena evaluates multimodal agent memory through action–world interaction, diagnosing write, maintain, retrieve, and use stages under lifelong evolution and agentic execution protocols (Liu et al., 2026).

It is important to distinguish three notions that are easy to conflate. Tracking changing state asks whether the system recovers the correct value at a time or checkpoint. Profile or lifecycle consistency asks whether personal attributes, stage metrics, or write/use pipelines remain coherent for a user or agent trajectory. Cross-query world-history consistency asks whether interdependent answers about current state, historical state, transitions, relations, and provenance jointly satisfy constraints induced by one shared multi-entity gold history.

Table 6: Closest related benchmarks and adjacent consistency work. Cells use Y / P / N for whether the feature is a primary evaluation object as operationalized in WorldConsistMem.

Work	LTM	Evol.	Bundles	Gold hist.	Trans.	Prov.	CQ constr.	Strict BCR	Partial CSR	Primary metric
LongMemEval	Y	P	N	P	P	N	N	N	N	Ability Acc
LongMemEval-V2	Y	Y	N	P	P	N	N	N	N	Acc (+ latency)
MemoryAgentBench	Y	P	N	P	P	N	N	N	N	Competency Acc
EverMemBench	Y	Y	N	P	P	N	N	N	N	Recall / profile Acc
EvoMemBench	Y	P	N	P	P	N	N	N	N	Task Acc
MemConflict	Y	P	N	P	P	P	P	N	P	Conflict Acc + retrieval
DynamicMem	Y	Y	P	Y	P	N	P	N	P	Checkpoint Acc
WorldMemArena	Y	Y	P	P	P	N	N	N	P	Lifecycle Acc
SetCons metrics	N	N	Y	P	P	N	Y	Y	P	SetCons / satisfiability
LogicVault	N	N	Y	P	P	N	Y	Y	P	Logical / SMT
WorldConsistMem	Y	Y	Y	Y	Y	Y	Y	Y	Y	Acc+BCR+CSR

C.3 Conflict, temporal validity, and provenance

MemConflict evaluates long-term memory systems under dynamic, static, and conditional memory conflicts, combining black-box answer correctness with white-box retrieval and ranking diagnostics, and showing that answer correctness can diverge from retrieval fitness (Tao et al., 2026). These lines diagnose conflict handling, version validity, and temporal updates at the query, retrieval, or competency level. The WorldConsistMem contrast is evaluation grain and joint structure rather than denial of those pathologies: a conflict- or update-correct single answer can still participate in an inconsistent bundle.

C.4 Cross-query logical consistency

Salla et al. (2026) study case-file logical consistency across interdependent queries and introduce set-level metrics including Case Satisfiability and SetConsRate. Chaudhry (2026) maintain persistent symbolic belief states verified with SMT-style checking and release LogicBench-Cross. Machine-checkable constraint evaluation over multi-query answer sets is therefore not universally new. WorldConsistMem’s residual is application domain and constraint origin: persistent memory systems over evolving multi-entity worlds with world-history-derived constraints.

D Fairness Caveats

Comparisons are informative but not perfectly matched:

- Internal structure differs (flat list vs multi-store H0).
- Capacity accounting charges H0 metadata (graph, provenance, indices) against the same logical-record budgets used for flat stores; absolute budgets therefore tax structured metadata.
- Retrieval budgets (k vs store-routed packs) are matched numerically where stated, but lexical BM25 is not identical to H0’s routing.
- Deterministic vs LLM readers are separate conditions; symbolic Acc/BCR must not be pooled with Qwen Acc/BCR.
- Compact flat can retain focal lifecycle events by dropping padding, which can yield very high symbolic Acc/BCR at medium budgets; this is a fairness caveat for matched-budget interpretation, not a general ranking of store designs.

E Additional Figures and Tables

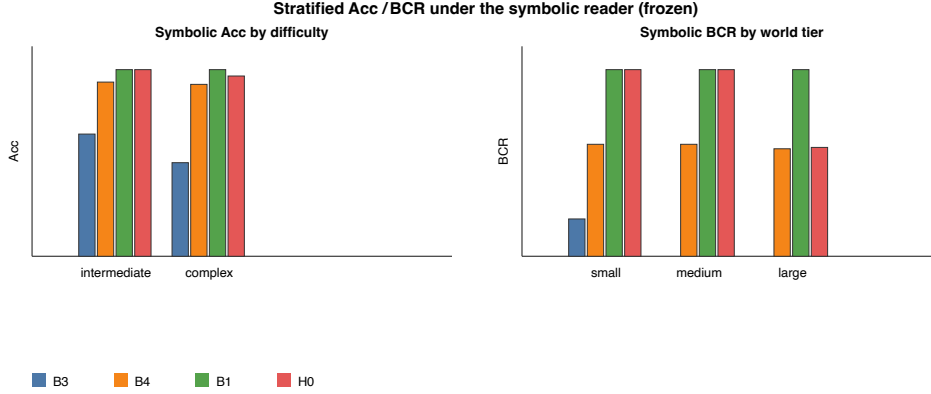


Figure 5: Symbolic Acc by difficulty and BCR by world tier.

Table 7: Constraint-family violation rates (headline systems).

Family	Symbolic				Qwen		
	B3	B4	B1	H0	B1/B3	B4	H0
uniqueness	0.156	0.000	0.000	0.000	0.219	0.177	0.000
temporal	0.406	0.000	0.000	0.156	0.354	0.323	0.281
transition	0.375	0.120	0.000	0.000	1.000	1.000	1.000
relational	0.562	0.343	0.000	0.000	0.625	0.625	0.625
provenance	0.625	0.057	0.000	0.000	0.938	0.724	1.000
multi_hop	0.344	0.124	0.000	0.000	0.406	0.406	0.406
contradiction	0.000	0.000	0.000	0.000	0.052	0.073	0.188

Table 8: Query-family Acc under Qwen (H4; frozen; three-decimal display).

Family	B1	B3	B4	H0
current_state	0.010	0.010	0.083	0.542
historical_state	0.162	0.162	0.089	0.547
transition	0.000	0.000	0.000	0.000
multi_hop	0.000	0.000	0.052	0.000
provenance	0.188	0.167	0.417	0.000
temporal_order	0.128	0.128	0.205	0.308
contradiction	0.083	0.083	0.167	0.000
counterfactual	0.000	0.000	0.000	0.000
conflict_validity	0.462	0.462	0.462	0.000

Table 9: Evaluation coverage and hard limits (frozen).

Dimension	Coverage
Paper shape	Benchmark + evaluation methodology
Symbolic reader	Full scaled (102 / 1,088 / 7,140)
LLM reader	Qwen2.5-3B-Instruct-4bit; H4 subset only
Phi / cloud models	Not run for Option A
Architecture superiority	Unsupported under Qwen (all BCR = 0)
Ecological validity	Not claimed (synthetic worlds)
Primary consistency metric	BCR (strict); CSR complementary